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Sync Manager, CDK, and the Desktop

Desktop implications of Palm OS® Cobalt

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What's new?

Now 3 types of handheld databases

- Classic (Palm OS® Garnet and earlier)
  - SyncMgr.h
  - Designed to work with existing APIs
  - Still has 64k (UInt16) record size limitation
  - Unique by name, separate namespace
- Extended (Palm OS® Cobalt)
  - SyncDm.h
  - Functionally equivalent to classic, with more verbose APIs
  - Record size limit now UInt32
  - Designed to support partial reads/writes
  - Unique by name/creator, separate namespace
- Schema (Palm OS® Cobalt)
  - SyncDb.h
  - Strongly typed data
  - Better sync bookkeeping (no more slow syncs)
  - Bulk APIs
  - Unique by name/creator, separate namespace



What's new?

New conduit registration scheme

- New "System" level conduits
  - Now have dynamic conduit configurations per OS user
    - Windows user 1 can use PalmSource Desktop conduits
    - Windows user 2 can use Outlook conduits
  - Existing conduit scheme now called "User" conduits
    - Works well with existing conduit installers as they customize as opposed to replace the existing PalmSource Desktop conduits



What's new?

New conduit registration scheme

- New folder based conduit registration
  - Conduit registration info retrieved through "GetConduitInfo" entry point
  - No more API calls or utilities to get conduits to run
  - Makes conduit installers much simpler
  - Hitting first break point much easier
  - Two conduit folder per OS user
    - C:\Documents and Settings\All Users\Application Data\HotSync\Conduits
    - C:\Documents and Settings\Cole\Application Data\HotSync\Conduits
  - HotSync® refreshes conduits dynamically



What's new?

New conduit registration scheme

- New optional conduit behavior
  - Can opt out of requirement that Creator ID exists on handheld
    - No more "calc"!
    - Still requires a Creator ID for unique identification
    - Use wisely, conduit could create zombie databases (data with no associated application)
  - Can opt out of HotSync displaying conduit in *Custom…* list
    - Neat trick for supporting multiple versions of your application without confusing your end user
  - Can opt out of HotSync displaying conduit information in progress dialog
    - Allows conduit to make run time decisions without alerting the end user



What's new?

COM APIs

- New Extended and Schema database APIs available in COM
  - Like before, COM adds intuitive object model to database access
  - Desktop now ships with COM APIs installed
  - Works with both VB and VB.net, as well as any other COM enabled language
  - New Schema database access especially easy with column names
    - Demo at end

What's new?

Trusted Sync

- Can register new handheld Sync Clients besides HotSync®
  - Sync clients have separate access privileges to databases
- Only "Trusted Desktops" are granted access to secure databases by HotSync®
  - User must grant trust to a desktop
  - Secure databases are only backed up on trusted Desktops
  - Defaults apply if security is "none"

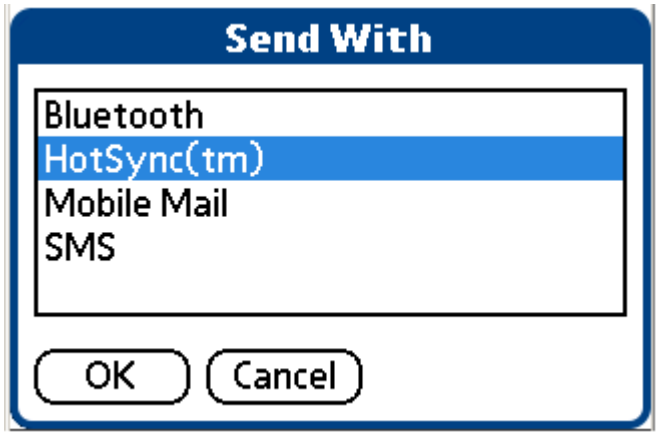




What's new?

New HotSync® Exchange allows data import/export without the creation of a conduit

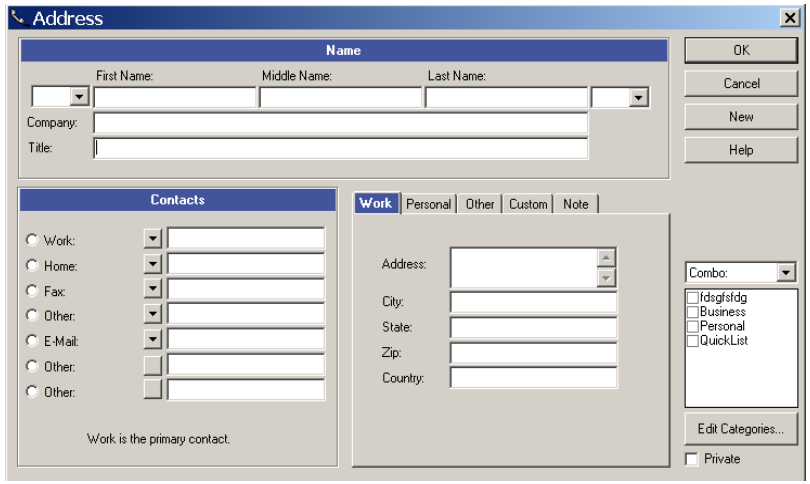
- Leverages handheld Exchange Manager
  - Simply "Send With HotSync" to get data onto the Desktop
    - Shows up in user's directory on Desktop
  - Simply use Desktop Install Tool to send data to the handheld
    - Same as if you beamed it



What's new?

Palm Desktop 6.0

- Address and Datebook added fields (Outlook compatibility)
- 256 Categories per application + multiple category membership
- Time Zone support in Datebook
- Backward compatibility with devices based on Palm OS 3.5
- Support for schema databases in all PIMs
- Desktop file format has changed





What's new?

Palm Desktop 6.0

- Compatible with: Windows 98SE, Me, 2000, XP
  - All user data converted to Desktop 6.0
  - Data conversion occurs at install time or at sync time
  - Mapping of contact fields occurs automatically

	Palm OS® 3.5 to Palm OS® Garnet	Palm OS® Cobalt
4.x Palm Desktop	**Yes**	**No** Palm OS Cobalt device cannot sync with pre-Desktop 6.0 due to sync and security constraints.
6.0 Palm Desktop	**Yes** Few restrictions apply. Only a subset of the data is synced to the device.	**Yes**

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The details: How to update your existing conduit

The upgrade from Classic to Extended is straightforward

- Every existing Classic API has a corresponding Extended API
  - Don't mix and match APIs
  - Take into account new record size limit
  - Take into account different endianess and structure packing
  - Don't assume unique database name anymore
  - Look at updated Generic Conduit sample as an example
    - Uses new Extended APIs
    - Uses XML as desktop file format
- MFC no longer in HotSync
  - No more debug MFC version issues
  - No more separate version of HotSync for Desktop and CDK
  - No change required for existing non-extension MFC conduits.
- Classic PIM applications not on Palm OS Cobalt devices
  - Data is migrated to new PIM Schema databases
  - Must use new desktop schema API to access PIM data (unlike PACE trick)



The details: How to update your existing conduit

Easier to support multiple conduits for multiple versions of your application

- No need to update your conduit if you don't update your application
- Recommend you have separate creator IDs for different types of databases
  - PalmSource uses "memo" and "mems" for the different types of MemoPad
- Check if the handheld is capable of running your application
  - "memo" opts out of Palm OS Cobalt devices and "mems" opts out of Palm OS Garnet and earlier devices
  - See conduit samples for example implementation



The details: Schema Databases

Schema database APIs differ greatly from Classic/Extended APIs

- Memory allocation
  - The API allocates the memory, the caller releases it
    - SyncDbReleaseStorage
  - Greatly reduces the number of round trips
- Bulk operations
  - Bulk read/write
    - SyncDbReadRows
    - SyncDbWriteRows
  - Bulk create/delete
    - SyncDbCreateRows
    - SyncDbDeleteRows
- Extensible
  - Add your own column to do ID mapping, add a birthday field to AddressBook, etc.
    - SyncDbAddColumns



The details: Schema Databases

Sync granularity much finer

- Not just a “dirty” bit for data
  - Schema databases keep a sync clock that indicates when it was last synchronized
  - Each desktop knows it’s clock value and gets only the updates it needs
- Separate bookkeeping for Schemas, Categories, and Rows.
  - Use SyncDbGetSyncMode to find out the state of the handheld
    - Fast Sync
    - Fast Sync with deletes
    - Slow Sync (now exceptional behavior)



The details: Schema Databases

```
// How to read the Sync Mode
// error checking assumed…
HSByte hHandle;
UInt32 dwID;
CmConvertStringToCreatorID(“mems”, &dwID);
SyncDbOpenDatabase(“MemoDBS”, dwID, dmModeReadWrite, dbShareNone, &hHandle);
DbSyncMode syncMode;
SyncDbGetSyncMode(hHandle, &syncMode, 0, NULL);
if (syncMode.row.type == dbNoChange)
        //  no modified rows
if (syncMode.row.type == dbFastSync)
        // must compare modified rows
        if (syncMode.row.flags == dbDeletesPurged)
                // must compare row IDs for missings IDs
if (syncMode.row.type == dbSlowSync)
        // must compare all rows (rare)

// do the same for syncMode.category and syncMode.schema
```



The details: Schema Databases

```
// How to read modified rows
// error checking still assumed…
UInt32 iRows;
DbRowData* pRowArray, pCurrentRow;
SyncDbReadModifiedRows(hHandle,“MemoDBSchema”,0,NULL DbMatchAny,&iRows, &pRowArray)
for (UInt32 iRow = 0; iRow < iRows; iRow++)
    pCurrentRow = pRowArray[iRow];
    if (pCurrentRow->changeFlags == dbDataModified)
            UInt32 iColumns = pCurrentRow->numColumns;
            for (UInt32 iColumn = 0; iColumn < iColumns; iColumn++)
                  DbColumnValue* pColumnValue = pCurrentRow>pColumnValues[iColumn];
                  // try to find the pCurrentRow->RowID in the other database and sync
                  // pColumnValue->columnData to the same pColumnValue->columnID

    // also check dbMembershipModified, dbRowDeleted, dbRowArchived

SyncDbReleaseStorage(pRowArray);
SyncDbCloseDatabase(hHandle, SYNC_CLOSE_DB_OPT_SYNCED_ALL_CHANGES);
```



The Demo: COM is cool

By Ravi Duggaraju (COM conduit guru)



Questions for the audience

What buzz words do you like to hear?

- N-Way Sync
- SyncML
- Macintosh support
- .net
- Eclipse support
- PalmSource Desktop integration SDK
- Java conduit support (cross platform)

?



Questions and Answers?

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http://www.palmos.com/dev/tech/conduits/

http://www.palmos.com/dev/support/forums/



palm OS® Cobalt

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